



Study Notes

# Monetary Aggregates

### Understanding terms related to Money

| Particular                                         | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Currency in Circulation                            | notes in circulation, rupee coins and small coins<br><i>Note - Currency with the public is arrived at after deducting cash with banks from total currency in circulation, as reported by RBI</i>                                                                                                                                                                                                                                                  |
| 'Bankers' deposits with RBI                        | balances maintained by banks in the current account with the RBI mainly for maintaining Cash Reserve Ratio (CRR) and as working funds for clearing adjustments                                                                                                                                                                                                                                                                                    |
| 'Other' Deposits with RBI                          | deposits from foreign central banks, multilateral institutions, financial institutions and sundry deposits net of IMF Account                                                                                                                                                                                                                                                                                                                     |
| Net RBI credit to Government                       | RBI's credit to Central as well as State Government i.e. <ul style="list-style-type: none"> <li>• WMA/overdrafts to govt.</li> <li>• RBI's holdings of G-Sec</li> <li>• holdings of rupee coins</li> <li>• <b>less</b> deposits of the concerned Government with RBI</li> </ul>                                                                                                                                                                   |
| Net bank credit to government                      | = Net RBI credit to the Government + Other banks' credit to the government                                                                                                                                                                                                                                                                                                                                                                        |
| RBI credit to the commercial sector                | <ul style="list-style-type: none"> <li>• investments in bonds/shares of financial institutions,</li> <li>• loans to them and</li> <li>• holdings of internal bills purchased and discounted</li> </ul>                                                                                                                                                                                                                                            |
| Bank credit to the commercial sector               | = RBI credit to the commercial sector + Other banks' credit to the commercial sector                                                                                                                                                                                                                                                                                                                                                              |
| Government's currency liabilities to the public    | <ul style="list-style-type: none"> <li>• rupee coins and small coins</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |
| RBI's net foreign assets                           | <ul style="list-style-type: none"> <li>• holdings of foreign currency assets and gold</li> </ul> <p><b>Note</b> – gold is valued at the end of the month at 90% of the daily average price quoted at London for the month. The rupee equivalent is determined on the basis of exchange rate prevailing on the last business day of the month</p> <p>Unrealised gains/ losses are adjusted to the Currency and Gold Revaluation Account (CGRA)</p> |
| Banking sector's Net foreign assets                | = RBI's net foreign assets + Other banks' foreign assets                                                                                                                                                                                                                                                                                                                                                                                          |
| Other Liabilities of RBI                           | <ul style="list-style-type: none"> <li>• internal reserves and provisions of RBI: <ul style="list-style-type: none"> <li>◦ Exchange Equalisation Account (EEA),</li> <li>◦ Currency and Gold Revaluation Account (CGRA)</li> <li>◦ Contingency Reserve</li> <li>◦ Asset Development Reserve</li> </ul> </li> </ul>                                                                                                                                |
| RBI's net nonmonetary liabilities                  | <ul style="list-style-type: none"> <li>• No monetary impact, like: <ul style="list-style-type: none"> <li>◦ Paid-up capital and reserves of RBI</li> <li>◦ Contribution to National Funds</li> <li>◦ RBI employees EPF and superannuation funds</li> <li>◦ Bills payable</li> <li>◦ IMF quota subscription, etc.</li> </ul> </li> </ul>                                                                                                           |
| Net non-monetary liabilities of the banking sector | = RBI's net non-monetary liabilities + Net non-monetary liabilities of other banks                                                                                                                                                                                                                                                                                                                                                                |

| Particular      | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Demand deposits | <p>Demand deposits are checkable deposits and include CASA accounts (Current Account, Saving Account)</p> <ul style="list-style-type: none"> <li>Inter-bank deposits are not taken into account, therefore Net demand deposits are considered for monetary aggregate compilations</li> </ul> <p><b>Note – Savings deposits are considered in two parts for monetary calculations:</b></p> <ul style="list-style-type: none"> <li><i>Demand liabilities portion – this refers to the freely withdrawable portion of the savings deposit</i></li> <li><i>Time liabilities portion – this effectively refers to that portion which has to be maintained as minimum balance and therefore is not as easily withdrawable (or not withdrawable only after paying penal interest)</i></li> </ul> |

## Monetary Aggregates

| Money Term                   | Also referred to as                                                                                                                                                             | Calculated as                                                                                                                                                                                                                                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M0<br>(compiled weekly)      | <ul style="list-style-type: none"> <li><b>Reserve Money</b></li> <li>High Powered Money</li> <li>Basic Money/Primary Money Monetary Base</li> <li>Central Bank Money</li> </ul> | <p>= Currency in circulation + Bankers' deposits with the RBI + 'Other' deposits with the RBI</p> <p><b>OR</b></p> <p>= Net RBI credit to the government + RBI credit to the commercial sector + RBI's claims on banks + RBI's net foreign assets + Government's currency liabilities to the public – RBI's net nonmonetary liabilities</p> |
| M1<br>(compiled fortnightly) | <ul style="list-style-type: none"> <li><b>Narrow Money</b></li> </ul>                                                                                                           | <p>= Currency with the public + Demand deposits with the banking system + 'Other' deposits with the RBI</p> <p><b>OR</b></p> <p>Currency with the Public + Current Deposits with the Banking System + Demand Liabilities Portion of Savings Deposits with the Banking System + 'Other' Deposits with the RBI</p>                            |
| M2                           |                                                                                                                                                                                 | = M1 + Savings deposits of post office savings banks                                                                                                                                                                                                                                                                                        |
| M3<br>(compiled fortnightly) | <ul style="list-style-type: none"> <li>Aggregate Monetary Resource</li> <li><b>Broad Money</b></li> <li>Money Supply</li> </ul>                                                 | <p>= M1 + Time deposits with the banking system</p> <p><b>OR</b></p> <p>= Net bank credit to the government + Bank credit to the commercial sector + Net foreign assets of the banking sector + Government's currency liabilities to the public – Net non-monetary liabilities of the banking sector</p>                                    |
| M4                           |                                                                                                                                                                                 | M3 + All deposits with post office savings banks (excluding National Savings Certificates).                                                                                                                                                                                                                                                 |

### Note:

- Currently, Narrow Money (M1) and Broad Money (M3) are relevant indicators of money supply in India
- The major distinction between the M1 (narrow money) and M3 (broad money) is with respect to the deposits with the banks. M3 includes the time deposits also in addition to the demand deposits with the banks.
- The distinction between M1/M3 and M2/M4 respectively is related to the Post Office Savings. Now, M2 and M4 are irrelevant as Post office Savings is mostly negligible after the rise and prominence of scheduled banks over the years. As such, the data related to Post office savings is also not frequently updated by RBI.

Liquidity Relation among the Monetary Aggregates

$M1 > M2 > M3 > M4$

- M1 is the most liquid form of money
- M4 is the least liquid form of money

**Money Multiplier**

- Money Multiplier is the ratio of the Narrow Money (M1) or the Broad Money (M3) to Reserve Money (M0)
- **Money multiplier (m) =  $1/\text{Reserve ratio requirement}$**
- **Money Supply =  $m * M0$**
- The amount money that banks generate with each unit of reserve money is through the money multiplier.
- As such, the supply of money is product of Money Multiplier (m) and the amount of high powered money or the reserve money.

**For example** let's say total deposit in banking system is Rs.100 crore and reserve ratio requirement is 5%. The banks can lend 95% of deposit i.e. Rs.95 crore. This Rs.95 crore that banks will lend to its customers will ultimately be deposited in another bank which can further lend 95% of that i.e. Rs.90.25 crore and cycle continues.

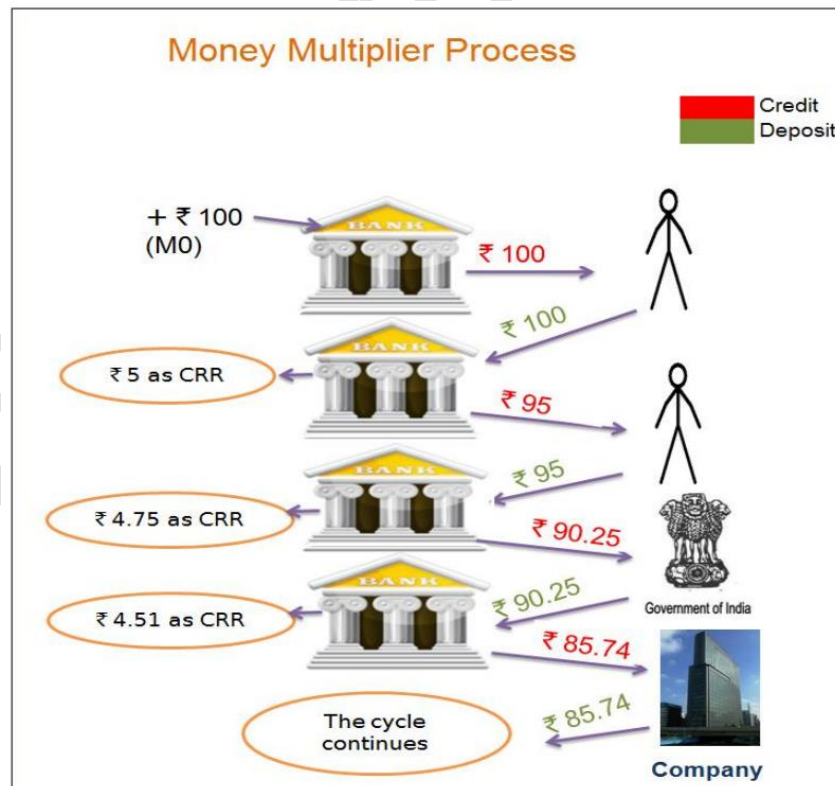


Image Source: IDFC Mutual Fund

### New Monetary Aggregates and Liquidity Aggregates

- In view of the changes in the Indian economy as well as the developments in monetary sector, working groups were set up periodically to review and refine the monetary aggregates.
- Three working groups were set up so far:
  - First Working Group on Money Supply (FWG) (1961),
  - the Second Working Group (SWG) (1977)
  - “Working Group on Money Supply: Analytics and Methodology of Compilation” (WGMS) (Chairman: Dr. Y.V. Reddy) (1998)
- NM3 is the **residency-based broad money aggregate** and L1, L2 and L3 are liquidity aggregates compiled on the recommendations of the Working Group on Money Supply (Chairman: Dr. Y.V. Reddy, 1998) as follows:

| Money Term       | Calculated as                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| NM1 (same as M1) | = Currency with the public + Demand deposits with the banking system + ‘Other’ deposits with the RBI.                                               |
| NM2              | = NM1 + Short-term time deposits of residents (including and up to the contractual maturity of one year).                                           |
| NM3              | = NM2+ Long-term time deposits of residents + Call/Term funding from financial institutions.                                                        |
| L1               | NM3 + All deposits with the post office savings banks (excluding National Savings Certificates).                                                    |
| L2               | L1 +Term deposits with term lending institutions and refinancing institutions (FIs) + Term borrowing by FIs + Certificates of deposit issued by FIs |
| L3               | = L2 + Public deposits of non-banking financial companies.                                                                                          |